

8

Investment Decision Rules

LEARNING OBJECTIVES

- Calculate Net Present Value
- Use the NPV rule to make investment decisions
- Understand alternative decision rules and their drawbacks
- Choose between mutually exclusive alternatives
- Evaluate projects with different lives
- Rank projects when a company's resources are limited so that it cannot take all positive-NPV projects

notation

CF_n	cash flow that arrives at date n	NPV	net present value
g	growth rate	PV	present value
IRR	internal rate of return	r	discount rate
$MIRR$	modified internal rate of return		

Managers are constantly faced with major investment decisions. In 2007, Apple decided to introduce its first mobile phone. Amazon.com entered the mobile phone business in 2014 and exited it in 2015. That same year, Tesla started offering the Model 3 for pre-sale, priced at less than half of its popular premium S-series. In 2016, Amazon.com decided to start flying its own cargo planes, leasing 20 Boeing 767 freighters. Starting in 2018, Netflix bet big on original content, allocating \$13 billion in that year and more than \$15 billion in 2019. How did Apple, Amazon, Tesla, and Netflix managers decide to commit substantial capital to these initiatives? In this chapter, we focus on the decision-making tools managers use to evaluate investment decisions. Examples of these decisions include new products, equipment purchases, and marketing campaigns.

We begin by introducing the *net present value (NPV) rule*. Although the NPV investment rule maximizes the value of the firm, some firms nevertheless use other techniques to evaluate investments and decide which projects to pursue. As a result, it is important to understand those techniques, so we will explain some commonly used alternative techniques—namely, the *payback rule* and the *internal rate of return (IRR) rule*. In each case, we define the decision rule and compare decisions based on this rule to decisions based on the NPV rule. We also illustrate the circumstances in which each of the alternative rules is likely to lead to bad investment decisions.

After establishing these rules in the context of a single, stand-alone project, we broaden our perspective to include evaluating multiple opportunities to select the best one. We conclude with a look at project selection when the firm faces limits on capital or managers' time.

8.1 The NPV Decision Rule

In the previous chapters, we learned how to convert between cash today and cash in the future using the discount rate. As long as we convert costs and benefits to the same point in time, we can use the Valuation Principle to make a decision. In practice, most corporations prefer to measure values in terms of their present value—that is, in terms of cash today. In this section, we apply the Valuation Principle to derive the concept of the *net present value* or *NPV*, which we can use to define the “golden rule” of financial decision making, the *NPV rule*.

Net Present Value

net present value (NPV) The difference between the present value of a project's or investment's benefits and the present value of its costs.

When the value of a cost or benefit is computed in terms of cash today, we refer to it as the present value (PV). Similarly, we define the **net present value (NPV)** of a project or investment as the difference between the present value of its benefits and the present value of its costs:

Net Present Value

$$NPV = PV(\text{Benefits}) - PV(\text{Costs}) \quad (8.1)$$

Let's consider a simple example. Suppose your firm is offered the following investment opportunity: In exchange for \$500 today, you will receive \$550 in one year. If the interest rate is 8% per year, then:

$$\begin{aligned} PV(\text{Benefit}) &= (\$550 \text{ in one year}) \div (\$1.08 \text{ in one year}/\$1 \text{ today}) \\ &= \$509.26 \text{ today} \end{aligned}$$

This PV is the amount you would need to put in the bank today to generate \$550 in one year ($\$509.26 \times 1.08 = \550). In other words, *the present value is the amount you need to invest at the current interest rate to recreate the cash flow*. We can think of this as the cash cost today of generating the cash flow ourselves.

Once the costs and benefits are in present value terms, we can compare them by computing the investment's NPV:

$$NPV = \$509.26 - \$500 = \$9.26 \text{ today}$$

The NPV is positive, so the benefits outweigh the costs, which means the firm should undertake this investment opportunity.

But what if you don't have the \$500 needed to cover the initial cost of the project? Does the project still have the same value? Because we computed the value using competitive market prices, it should not depend on your tastes or the amount of cash you have in the bank. If you don't have the \$500, suppose you borrow \$509.26 from the bank at the 8% interest rate and then take the project. What are your cash flows in this case?

Today: \$509.26 (loan) – \$500 (invested in the project) = \$9.26

In one year: \$550 (from project) – $\$509.26 \times 1.08$ (loan balance) = \$0

This transaction leaves you with exactly \$9.26 extra cash in your pocket today and no future net obligations. So taking the project is similar to having an extra \$9.26 in cash up front. Thus, the NPV expresses the value of an investment decision as an amount of cash received today. *As long as the NPV is positive, the decision increases the value of the firm and is a good decision regardless of your current cash needs or preferences regarding when to spend the money.*

EXAMPLE 8.1

The NPV Is
Equivalent to Cash
Today

PROBLEM

After saving \$1500 by waiting tables, you are about to buy a 50-inch TV. You notice that the store is offering a "one year same as cash" deal. You can take the TV home today and pay nothing until one year from now, when you will owe the store the \$1500 purchase price. If your savings account earns 5% per year, what is the NPV of this offer? Show that its NPV represents cash in your pocket.

SOLUTION

PLAN

You are getting something worth \$1500 today (the TV) and in exchange will need to pay \$1500 in one year. Think of it as getting back the \$1500 you thought you would have to spend today to get the TV. We treat it as a positive cash flow.

Cash flows:

Today	In one year
+\$1500	-\$1500

The discount rate for calculating the present value of the payment in one year is your interest rate of 5%. You need to compare the present value of the cost (\$1500 in one year) to the benefit today (a \$1500 TV).

EXECUTE

$$NPV = +1500 - \frac{1500}{(1.05)} = 1500 - 1428.57 = \$71.43$$

You could take \$1428.57 of the \$1500 you had saved for the TV and put it in your savings account. With interest, in one year it would grow to $1428.57 \times (1.05) = 1500$, enough to pay the store. The extra \$71.43 is money in your pocket to spend as you like (or put toward the speaker system for your new media room).

EVALUATE

By taking the delayed payment offer, we have extra net cash flows of \$71.43 today. If we put \$1428.57 in the bank, it will be just enough to offset our \$1500 obligation in the future. Therefore, this offer is equivalent to receiving \$71.43 today, without any future net obligations.

The NPV Decision Rule

As shown in Example 8.1, the Valuation Principle implies that we should undertake projects with a positive NPV because undertaking those projects increases wealth. That is, good projects are those for which the present value of the benefits exceeds the present value of the costs. As a result, the value of the firm increases and investors are wealthier. Projects with negative NPVs have costs that exceed their benefits. Accepting them is equivalent to losing money today.

We capture this logic in the **NPV Decision Rule**:

When making an investment decision, take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today.

NPV decision rule When choosing among investment alternatives, take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today.

Because NPV is expressed in terms of cash today, using the NPV decision rule is a simple way to apply the Valuation Principle. Decisions that increase wealth are superior to those that decrease wealth. We don't need to know anything about the investor's preferences to reach this conclusion. As long as we have correctly captured all the cash flows of a project and applied the appropriate discount rate, we can determine whether the project makes us wealthier. Being wealthier increases our options and makes us better off, whatever our preferences are.

A common way companies apply the NPV rule in financial practices is when deciding whether to accept or reject a project. Because rejecting the project generally has $NPV = 0$ (there are no new costs or benefits from not doing the project), the NPV decision rule implies that we should

- Accept positive-NPV projects; accepting them is equivalent to receiving their NPV in cash today, and
- Reject negative-NPV projects; accepting them would reduce the value of the firm, whereas rejecting them has no cost ($NPV = 0$).

If the NPV is exactly zero, then you will neither gain nor lose by accepting the project instead of rejecting it, which also has an NPV of zero. It is not a bad project because it does not reduce the firm's value, but it does not add value to the firm either.

CONCEPT CHECK

1. What is the NPV decision rule? How is it related to the Valuation Principle?
2. Why doesn't the NPV decision rule depend on the investor's preferences?

8.2 Using the NPV Rule

We continue our discussion of investment decision rules by considering a take-it-or-leave-it decision about a single, stand-alone project. By undertaking this project, the firm does not constrain its ability to take other projects. In the case of a stand-alone project, the alternatives we are considering are to accept or reject a project. The NPV rule then implies that we should compare the project's NPV to zero (the NPV of rejecting the project and doing nothing). Thus, we should accept the project if its NPV is positive.¹

Organizing the Cash Flows and Computing the NPV

Researchers at Fredrick's Feed and Farm have made a breakthrough. They believe they can produce a new, environmentally friendly fertilizer at a substantial cost saving over the company's existing line of fertilizer. The fertilizer will require a new plant that can be built immediately at a cost of \$81.6 million. Financial managers estimate that the benefits of the new fertilizer will be \$28 million per year, starting at the end of the first year and lasting for four years, as shown by the following timeline:

Year:	0	1	2	3	4
Cash flow (in millions):	-\$81.60	+\$28	+\$28	+\$28	+\$28

Thus, the cash flows are an immediate \$81.6 million outflow followed by an annuity inflow of \$28 million per year for four years. Therefore, given a discount rate r , the NPV of this project is:

¹Some prefer to state the rule that the project should be accepted if its NPV is non-negative. Since the firm should be indifferent at an NPV of exactly zero (you will neither gain nor lose by accepting the project instead of rejecting it), we prefer to state it this way. A zero NPV project is not a bad project because it does not reduce the firm's value, but it does not add value to the firm either.

$$NPV = -81.6 + \frac{28}{1+r} + \frac{28}{(1+r)^2} + \frac{28}{(1+r)^3} + \frac{28}{(1+r)^4} \quad (8.2)$$

We can also use the annuity formula from Chapter 4 to write the NPV as:

$$NPV = -81.6 + \frac{28}{r} \left(1 - \frac{1}{(1+r)^4} \right) \quad (8.3)$$

To apply the NPV rule, we need to know the cost of capital. The financial managers responsible for this project estimate a cost of capital of 10% per year. If we replace r in Eq. 8.2 or 8.3 with the project's cost of capital of 10%, we get an NPV of \$7.2 million, which is positive. Recall that a net present value tells us the present value of the benefits (positive cash flows) net of the costs (negative cash flows) of the project. By putting everything into present values, it puts all the costs and benefits on an equal footing for comparison. In this case, the benefits outweigh the costs by \$7.2 million in present value. The NPV investment rule indicates that by making the investment, Fredrick's will increase the value of the firm today by \$7.2 million, so Fredrick's should undertake this project.

The NPV Profile

The NPV of the project depends on its appropriate cost of capital. Often, there may be some uncertainty regarding the project's cost of capital. In that case, it is helpful to compute an **NPV profile**, which graphs the project's NPV over a range of discount rates. It is easiest to prepare the NPV profile using a spreadsheet such as Excel. We simply repeat our calculation of the NPV above using a range of different discount rates instead of only 10%. Figure 8.1 presents the NPV profile for Fredrick's project by plotting the NPV as a function of the discount rate, r .²

NPV profile A graph of a project's NPV over a range of discount rates.

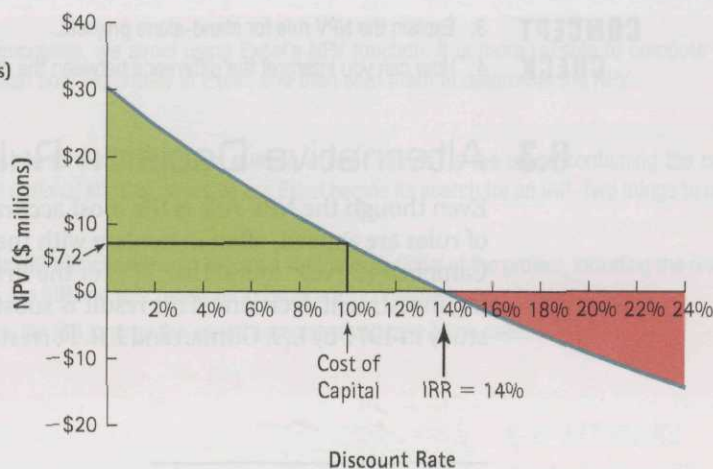
FIGURE 8.1
NPV of Fredrick's
New Project

The graph in Panel (b) shows the NPV as a function of the discount rate based on the data in Panel (a). The NPV is positive, represented by the green-shaded area, only for discount rates that are less than 14%, the *internal rate of return (IRR)*. Given the cost of capital of 10%, the project has a positive NPV of \$7.2 million. The red-shaded area indicates discount rates above the 14% IRR with negative NPVs.

Panel (a)

Discount Rate	NPV (\$ millions)
0%	\$30.4
2%	\$25.0
4%	\$20.0
6%	\$15.4
8%	\$11.1
10%	\$7.2
12%	\$3.4
14%	\$0.0
16%	-\$3.3
18%	-\$6.3
20%	-\$9.1
22%	-\$11.8
24%	-\$14.3

Panel (b)



²In the appendix to this chapter, we show you how to create an NPV profile in Excel.

internal rate of return (IRR) The interest rate that sets the net present value of the cash flows equal to zero.

Notice that the NPV is positive only for discount rates that are less than 14% (the green shaded area in Figure 8.1). Referring to the graph and the accompanying data table, we see that at 14% the NPV is zero. The discount rate that sets the net present value of the cash flows equal to zero is an investment's **internal rate of return (IRR)**. Thus, by constructing the NPV profile, we have determined that Fredrick's project has an IRR of 14%. We can also compute the IRR without graphing the NPV by using a financial calculator or a spreadsheet's IRR function (see the appendix to Chapter 4. The calculator instructions for solving for the rate of return can be used to solve for the IRR).

	N	I/Y	PV	PMT	FV
Given:	4		-81.6	28	0
Solve for:		14			
Excel Formula: =RATE(NPER,PMT,PV,FV)=RATE(4,28,-81.6,0)					

Measuring Sensitivity with IRR

In our Fredrick's example, the firm's managers provided the cost of capital. If you are unsure of your cost of capital estimate, it is important to determine how sensitive your analysis is to errors in this estimate. The IRR can provide this information. For Fredrick's, if the cost of capital estimate is more than the 14% IRR, the NPV will be negative (see the red shaded area in Figure 8.1). Therefore, as long as our estimate of the cost of capital of 10% is within 4% of the true cost of capital, our decision to accept the project is correct. In general, the difference between the cost of capital and the IRR tells us the amount of estimation error in the cost of capital estimate that can exist without altering the original decision.

Alternative Rules Versus the NPV Rule

The NPV rule indicates that Fredrick's should undertake the investment in fertilizer technology. As we evaluate alternative rules for project selection in the subsequent sections, keep in mind that sometimes other investment rules may give the same answer as the NPV rule, but at other times they may disagree. When the rules conflict, always base your decision on the NPV rule, which is the most accurate and reliable decision rule.

CONCEPT CHECK

3. Explain the NPV rule for stand-alone projects.
4. How can you interpret the difference between the cost of capital and the IRR?

8.3 Alternative Decision Rules

Even though the NPV rule is the most accurate and reliable rule, in practice a wide variety of rules are applied, often in tandem with the NPV rule. In a 2022 study, John Graham and Campbell Harvey³ found that 77% of the firms they surveyed used the NPV rule for making investment decisions. This result is substantially different from that found in a similar study in 1977 by L. J. Gitman and J. R. Forrester,⁴ who found that only 10% of firms used the

³John Graham, "Presidential Address: Corporate Finance and Reality," *Journal of Finance* 77 (2022): 1975–2049.

⁴L. J. Gitman and J. R. Forrester, Jr., "A Survey of Capital Budgeting Techniques Used by Major U.S. Firms," *Financial Management* 6 (1977): 66–71.